HH-I-528b
May 26, 1966

SUPERSEDING
Fed. Spec. HH-I-528a
October 3, 1951

FEDERAL SPECIFICATION

INSULATION BATTS AND BLANKETS, THERMAL (VEGETABLE FIBER)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for vegetable fiber insulation.

1.2 Classification. Insulation covered by this specification shall be of one grade and of the following classes (see 6.2):

1—Without membrane facing.
2—With membrane facing on one face.
3—With enveloping membrane.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-B-790—Building Paper, Vegetable Fiber: (Kraft, Waterproofed, Water Repellent and Fire Resistant).
UU-P-31—Paper; General Specifications and Methods of Testing.
PPP-B-636—Box, Fiberboard.

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials (ASTM) Publications:

C 167—Thickness and Density of Blanket or Batt-Type Thermal Insulating Weight.
C 177—Thermal Conductivity of Materials by means of the Guarded Hot Plate.

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E 84—Surface Burning Characteristics of Building Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa., 19103.)

Official Classification Committee (OCC) Publication:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Ave. at 33rd St., New York, N. Y., 10016.)

American Trucking Associations, Inc. (ATA) Publication:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P St., N.W., Washington, D. C., 20036.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. The batt and blanket insulation shall be composed of firmly matted, well-cleaned, new cotton fibers, and shall have reduced flammability, as specified in 3.8. All materials shall be free from objectionable odor.

3.2 Construction. The insulation shall be fabricated in a flexible form, and to approximately uniform thicknesses. They shall be with or without membrane facing on both principal faces.

3.2.1 *Membrane facings.* When membrane facings are required, the facings shall be waterproof, vapor-resistant, or vapor-permeable as specified. Vapor-resistant and vapor-permeable membranes shall conform to grades A and D, respectively of UU-B-790.

3.2.2 *Class* 1. The batts shall be without membrane facing on both principal faces.

3.2.3 *Class* 2. The blankets shall be provided with a membrane facing attached to one principal face. The facing shall extend at least one inch beyond the blankets on both longitudinal edges so as to form nailing flanges. Unless otherwise specified (see 6.2), the facing shall be waterproof and vapor-resistant, exclusive of the nailing flanges.

3.2.4 *Class* 3. The blankets shall be provided with enveloping membranes. The nailing flanges shall extend one inch or more beyond the longitudinal edges of the blankets. Unless otherwise specified (see 6.2), the facing on one side shall be both waterproof and vapor-resistant, and the other face shall be vapor-permeable.

3.3 Standard product. Except as specified herein, the insulation and its component parts shall be a standard commercial product of the manufacturer or his suppliers. The items furnished shall be new and unused.

3.4 Weight. The insulation material shall weigh not less than 0.052 pound per square foot for each inch of nominal thickness when the weight is determined in accordance with 4.4.2.

3.5 Dimensions. The batts and blankets shall be of specified length and width and shall be of nominal thicknesses of 3/4, 1, 1-1/2, 2, 2-1/2, 3, 3-1/2, and 4 inches, as specified (see 6.2). A minus tolerance in thickness of 1/8 inch per inch of nominal thickness will be permitted when tested as specified in 4.4.2. The batts and blankets may be ordered by thermal resistance (R) factors. For a 1 inch thickness the R factor is 3.33.

3.6 Flexibility. The batts and blankets shall be sufficiently flexible to bend around a rod 2 inches in diameter without injuring the cotton insulation or membrane facings.

3.7 Thermal conductivity. The thermal conductivity of the insulation batts and blankets shall not exceed 0.30 B.t.u. per hour per square foot for a temperature gradient of 1°F. per inch of thickness at a

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mean temperature of 75°F. when tested in accordance with 4.4.3.

3.8 **Flammability.** The insulation, exclusive of membrane facings, shall be uniformly and thoroughly treated with compounds to render it less flammable. When subjected to the test described in 4.4.4, the insulation shall have a flamespread rating of not more than 25.

3.9 **Workmanship.** The quality of workmanship shall meet the standards prevalent among manufacturers who normally produce insulation batts and blankets of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Inspection lot.** All insulation of the same class offered to the Government under a specific contract shall be considered a lot for purposes of inspection.

4.3 **Sampling.**

4.3.1 *Sampling for examination.* A random sample of batts and blankets shall be selected from each lot in accordance with MIL-STD-105 at inspection level I. In terms of defects per 100 units, the acceptable quality levels (AQL) shall be 1.5 for major defects and 4.0 for minor defects.

4.3.2 *Sampling for tests.* A random sample of batts and blankets shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level S-2. The AQLs shall be based on 4.0 percent defective units.

4.3.3 *Inspection.* Each unit in the sample for examination 4.3.1, shall be inspected as applicable for size, thickness, smoothness, appearance, packing, and marking. Standards for rejection shall be based on a standard sample.

4.4 Tests.

4.4.1 *Conditioning of test samples.* Unless otherwise specified (see 6.2), samples shall be conditioned for testing by exposure for at least 12 hours to the prevailing room atmosphere.

4.4.2 *Thickness and weight.* The thickness and weight of the cotton insulation shall be determined in accordance with ASTM C 167. The average thickness of representative specimen taken from each of the samples shall not fall below the specified thickness by more than 1/8 inch for each inch of nominal thickness.

4.4.3 *Thermal conductivity.* The thermal conductivity of the insulation shall be determined in accordance with ASTM C 177 except that the test specimen shall be conditioned as specified under 4.4.1.

4.4.4 *Flammability.* The flammability of the insulation shall be determined in accordance with ASTM E 84, unless otherwise specified (see 6.2).

4.4.5 *Water-vapor permeability.* The water-vapor permeability of the membrane shall be determined in accordance with method 182 of UU-P-31.

4.5 **Inspection of preparation for delivery.** The packing and marking of the insulation shall be inspected to determine conformance with the applicable requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

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5.1.1 *Level A.* The insulation shall be packed in boxes conforming to PPP-B-636 (class weather-resistant) and waterproof sealed with tape in accordance with the appendix thereto.

5.1.2 *Level B.* The insulation shall be packed in boxes conforming to PPP-B-636 (class domestic). Closure shall be in accordance with method I of the appendix to the box specification.

5.1.3 *Level C.* The complete equipment shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.2 **Marking (see 6.2).**

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** The insulation covered by this specification is intended for use in walls, ceilings, and floors.

6.2 **Ordering data.** Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Class required (see 1.2).

(*c*) When facings shall not be waterproof and vapor-resistant (see 3.2.3).

(*d*) When facings shall not be waterproof, vapor-resistant, and vapor-permeable (see 3.2.4).

(*e*) Dimensions required (see 3.5).

(*f*) Conditioning of test samples, if different (see 4.4.1).

(*g*) Test for flammability (see 4.4.4).

(*h*) Level of packing required (see 5.1).

(*i*) Marking required (see 5.2).

MILITARY CUSTODIANS:

Army—MO
Navy—YD
Air Force—69

Review activities:

Army—CE
Navy—YD
Air Force—69

User activities:

Army—MO
Navy—SH

Preparing activity:

Navy—YD

CIVIL AGENCIES INTEREST

AGR
HEW
JUS

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization documents.

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Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

HH-I-528b: Insulation Batts and Blankets, Thermal, (Vegetable Fiber)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426 IA 0101-807-2000 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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